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classmate

Date _____

Page _____

1) struct
freelist nodes
kmemctl
spinlock lock
int use_lock
freelist

lalloc
semes
head

↓
kinit & kinit 2
add pages
to freelist
↓
before
adds
page
to head

Page 1
struct run
(next)

5KB

Addr: 0x00100000

Page 2
struct run
(next)

Addr: 0x00200000

Page N
struct run
(next, (NULL))

Bhugale
04-03-23

2) Page directory

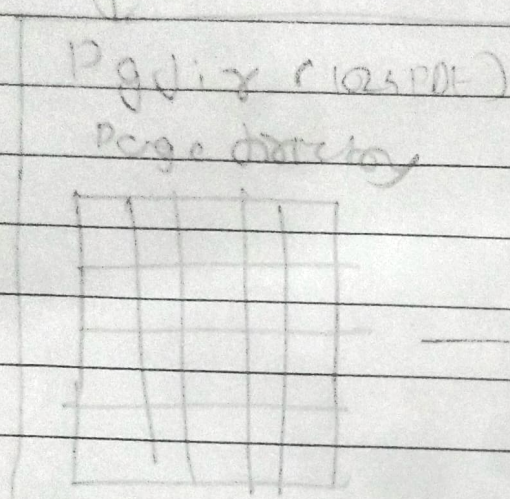
PTE - Page table entry

only

PDE - Page directory entry

only

page table & pgdir also switch from 1000
pgdir into CR3



→ PDE [KERNBASE]

Page table for KERNBASE

PDE [0]
Unused

PTE 0

Physical address
→ 0x00000000

PTE 1

Physical address
→ 0x01000000

PTE 2

Physical address
→ 0x02000000

Bhaskar
04-08-25